

Computer Networks

(IT25-424)

Course outline:

Network Topologies and Models, Network Layering Concepts and Protocols, Internet Protocol (IP) and Associated Control Protocols. End-To-End Protocols, With TCP and UDP as Examples, Addressing schemes at link layer, network layer and transport layer, Transmission Media and characteristics, Switching Techniques, Channel Access Techniques, MAC, Routing Protocols and Multicast, Overview of Application Layer Protocols (HTTP, FTP, SMTP etc.), Multimedia Protocols (RTP, RTSP, RTCP) , Security Mechanisms and Services, Concepts of Symmetric and Asymmetric Cryptography, Digital Signature, Convergence of communication networks

Learning Material/References:

- * Data Communications & Networking by Behroz Forouzan
- * Data and Computer Communications by William Stallings
- * Computer Networks, 4th Edition by Andrew S Tanenbaum